1 Two-way clustering

Cluster the standard error at time level vs. two-way clustering at time and cohort level.

Cohort is defined by individuals birth year, i.e., all individuals who are born in the same year belongs to the same cohort.

Cluster at time level:

- Allow for correlation of unkown form in residusals across time periods.

Cluster at time and cohort level (two-way clustering):

- Allow for correlation of unkown form in residusals within the same cohort across time periods (the 1980 cohort in 2015 and 2020) and for correlation across cohorts within the same time period (everyone in 2020).
- It also allows correlation of residuals within the exact same time-cohort cell (the 1980 cohort in 2020)

Cluster at time-cohort level:

- It ONLY allows correlation of residuals within the exact same time-cohort cell (the 1980 cohort in 2020)
- It assumes the correlation across cohorts within the same time period to be ZERO (cohort 1980 and cohort 1990 in 2020 share no common shocks. This is very restrictive!)
- It assumes the correlation within the same cohort across time periods to be ZERO (cohort 1980 in 2020 is completely independent of the 1980 cohort in 2021. This is very restrictive!)
- **In short: Two-way clustering is much more general and conservative, whereas clustering at the time-cohort cell level is much more restrictive.**

Number of Clusters:

- If you have 60 time periods and 50 cohorts,

- Two-way clustering relies on $G_{time} = 60$ and $G_{cohort} = 50$
- Time-cohort clustering creates $60 \times 50 = 3000$ clusters.

Magnitudes of SE:

- Clustering at the time-cohort cell level will almost always yield much smaller standard errors than two-way clustering. This is because the time-cohort level creates thousands of tiny clusters and falsely assumes that errors across different years or across different cohorts are completely independent.

Interpretation of SE under alternative clutering methods:

Two-way clustering computes the variance-covariance matrix as:

$$\widehat{Var}_{two-way} = \widehat{Var}_{time} + \widehat{Var}_{cohort} - \widehat{Var}_{time \times cohort}.$$

Consider results in Table 1, The SE is 0.0054 when it is clustered at time level. It is 50% larger than the SE clustered at the cohort level (0.0036). The two-way clustered SE is 0.006. It suggests that

- Time-series shocks are the dominant source of error dependence
  - The primary unobserved correlation in your data comes from aggregate macroeconomic shocks at time level (e.g., business cycles, sudden CPI releases, monetary policy announcements) that hit all households at the same time, regardless of their age.
  - Clustering only at the cohort level would severely understate your standard errors, leading to inflated t-statistics and false positives.
- Cohort persistence provides an incremental, non-trivial variance contribution
  - Moving from $SE_{time} = 0.0054$ to $SE_{two-way} = 0.006$ increases the SE by roughly 11% (and the variance by about 23%).
  - Economic interpretation: While aggregate time shocks dominate, individuals born in the same year still share persistent unobserved characteristics (e.g., formative lifetime economic experiences, life-cycle stage). Two-way clustering successfully absorbs this extra layer of dependence.

Variance Decomposition In Eq. (1),

$$\widehat{Var}_{time} = 0.0054^2 = 0.00002916$$
$$\widehat{Var}_{cohort} = 0.0036^2 = 0.00001296$$
$$\widehat{Var}_{two-way} = 0.0060^2 = 0.00003600$$

Hence,

$$\widehat{Var}_{time\times cohort} = \widehat{Var}_{time} + \widehat{Var}_{cohort} - \widehat{Var}_{two-way} = 0.00000612$$

$$SE_{time\times cohort} = \sqrt{\widehat{Var}_{time\times cohort}} \approx 0.00247$$

This reveals the hierachy of SEs:

$$SE_{time\times cohort} < SE_{cohort} < SE_{time} < SE_{two-way}$$

This hierarchy confirms that your error structure is well-behaved: standard errors grow as you allow for broader and more general structures of cross-sectional and temporal correlation.

Table 1: Alternative Clustering

	(1)	(2)	(3)
τ^{1yr}	0.0468***	0.0468***	0.0468***
	(0.006)	(0.0036)	(0.0054)
No. of Obs.	188,073	188,073	188,073
Pseudo-R-squared	0.2057	0.2057	0.2057
Clustering	Time-Cohort	Cohort	Time